



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2025

Bachelor in Computer Applications
Course Title: Computer Fundamentals and Applications
Code No: CACS101
Semester: I

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2024

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Explain the fundamental components of a computer system and their functions. [5]
3. Define computer memory. Explain memory hierarchy. [1+4]
4. What is computer software? Differentiate between a program and software. [1+4]
5. What is an Operating System (OS)? Explain different types of OS and give examples of Open-Source OS. [1+3+1]
6. What is a Database? Explain its main functions and types with examples. [1+3+1]
7. What is browsing? What is meant by home page of a website? Explain any three Internet protocols. [1+1+3]
8. Write short notes on (Any Two): [2.5 + 2.5]
a) Multimedia Vs Hypermedia b) GUI Vs CUI c) Text Formatting Vs Paragraph Formatting

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Why computer networks play a vital role in the 21st century? Explain the role of different transmission media in the network design. Describe the functionalities of various network devices. [2+2+6]
10. a) Write function and perform following calculation in MS-Excel. [5]

Places	Target	Production	Achievement (%)	Grade
Dolakha	4800	6000	?	?
Lalitpur	5000	4500	?	?
Mustang	5000	5000	?	?
Average			?	

- i) Calculates Achievement in percentage. [1]
- ii) Calculate failure or achievement in average percentage. [1]
- iii) Calculate grade A+ for more than 100% achievement; A for 100% achievement; B for below 100% achievement. [3]
- b) Explain the following DOS commands with syntax. [5]
 - i) How do you protect files in DOS?
 - ii) What is the difference between RD and DEL commands?
 - iii) How can you copy a file from one directory to another in DOS?
 - iv) What is the use of TREE and LABEL commands?
 - v) How do you display the contents of a text file without opening it?
11. a) Difference between adobe ImageReady and Photoshop. What are the different selection tools available in PS? [2+3]
- a) What is PowerPoint? What is its purpose? Differentiate between Slide Sorter view and Slide show view of PowerPoint. [1+1+3]



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Bachelor in Computer Applications
Course Title: Society and Technology
Code No: CASO102
Semester: I

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2024

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. "Society is the wave of social relationship." Discuss.
3. Sketch the evolutionary stages of society.
4. Illustrate competition as a social process.
5. What are the major bases of social stratification in Nepal?
6. Discuss major types of cybercrimes in Nepal.
7. Describe the key components of research proposal.
8. Discuss in brief the types of probability sampling.

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Explain different stages of socialization with its implications.
10. Discuss the various consequences of the technological advancement in the societies.
11. What are the major accountabilities of computer professionals?



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2025

Bachelor in Computer Applications
Course Title: English I
Code No: CAEN 103
Semester: I

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2024

Candidates are required to answer the questions in their own words as far as practicable.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Explain the role of technical vocabulary in computing. Provide examples of specialized terms and their significance in professional communication.
3. Point out the key differences between machine language and high-level programming languages. How does this impact software development?
4. Discuss the importance of user interface design in computing. What are some key principles that contribute to an effective UI?
5. Define the concept of networking in computing. How do protocols such as TCP/IP facilitate communication between computers?
6. Describe the impact of artificial intelligence on modern computing. What are some ethical concerns associated with AI advancements?
7. Match the following words from 1-5 with their definitions from a-e.

i) Capture	a) light-sensitive substances on the surface of film
ii) Poster	b) device that uses compressed air to spray paint
iii) Retouch	c) succeed in representing on film
iv) Airbrush	d) alter by making minor changes
v) Emulsion	e) large printed picture used for advertising purposes
8. Form a new word using each of the following prefixes:
Un- non- in- dis- trans-

Group C

[2×10 = 20]

Attempt any TWO questions.

9. With the increasing availability of online services such as email, file transfer, and remote access, how do these services enhance communication and data exchange in professional environments, and what are some potential challenges, including security risks and technical limitations, that users might face while using them.
10. Write a job application letter for the post of IT officer at a US based IT company, also prepare suitable resume.
11. How does virtual reality simulate real-world environments through hardware and software components, and what are the advantages and limitations of using VR for training and professional development in industries such as healthcare and aviation? Additionally, what challenges do developers face in improving the realism and interactivity of VR systems, and how might future advancements in artificial intelligence and haptic feedback enhance the immersive experience?



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Bachelor in Computer Applications
Course Title: Mathematics I
Code No: CAMT104
Semester: I

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2024

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. In an examination, 27% of the students failed in Mathematics and 31% failed in statistics. If 6% of the students failed in both subjects, find the percentage of the students
 - a) failed in examination.
 - b) passed in both subjects.
3. a) If $A = [-3, 1)$ and $B = [-2, 4]$. Find $A - B$ with graphical representation in a real line. [2.5]
b) Rewrite $-3 < x < 4$ by using the modulus sign.
4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + 1$ and $g(x) = 3x - 1$,
find i) $\text{gof}(x)$ ii) $\text{fog}(x)$ [2 + 3]
5. Sum to n terms of the series $1.3 + 3.5 + 5.7 + \dots$
6. Prove that:
$$\begin{vmatrix} a+x & b & c \\ a & b+y & c \\ a & b & c+z \end{vmatrix} = xyz \left(1 + \frac{a}{x} + \frac{b}{y} + \frac{c}{z} \right).$$
7. Find the centre, vertices, eccentricity, foci and length of major axis of the ellipse: $25x^2 + 4y^2 = 100$.
8. The letters of the word 'STRANGE' are to be shuffled to form different words such that
 - a) the vowels always come together.
 - b) the vowels may occupy only the odd positions

Group C

Attempt any TWO questions.

[2×10 = 20]

9. If arithmetic mean, geometric mean and harmonic mean between two unequal positive numbers are A , G , H respectively, then prove that:
 - a) $(G.M.)^2 = A.M. \times H.M.$
 - b) $A.M. > G.M. > H.M.$
10. a) Define scalar product of two vectors in space. By vector method, in triangle ΔABC prove that
 $c = a \cos B + b \cos A$.
b) Find the area of the parallelogram determined by the vectors $\vec{i} + 2\vec{j} + 3\vec{k}$ and $-3\vec{i} - 2\vec{j} + \vec{k}$.
11. Find the equation of a circle passing through the points $(1, 2)$, $(3, 1)$ and $(-3, -1)$.



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Bachelor in Computer Applications
Course Title: Digital Logic
Code No: CACS 105
Semester: I

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2024

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Define analog and digital quantities. Subtract: $1010101.101 - 1000100.001$ using both 1's and 2's complement. [1 + 2 + 2]
3. Convert $(2594.68)_{10}$ into hexadecimal number. Design full subtractor with its block diagram, truth table, logic diagram and Boolean expression. [2+3]
4. What NAND and NOR gates are called universal gates? Express the Boolean function $F(A,B,C,D) = D(A'+B) + B'D$ into standard SOP and POS. [1+2+2]
5. Combinational circuit that generates 9's complement of a 3 input numbers and implementing it using 3 gates only. Design 1:8 DEMUX using 1:4 DEMUX and 1:2 DEMUX. [3+2]
6. Define state table. Design a combinational circuit using a ROM. The circuit accepts a 3-bit number and generates an output binary number equal to the square of the input number. [1+4]
7. Design Mod-11 synchronous counter. [5]
8. How flip flop differs from latch. Explain clocked SR flip flop with logic diagram, truth table, characteristic table and excitation table. [1+4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Differentiate between PAL and PLA. Design a combinational circuit with four inputs that represent a decimal digit in BCD and four output lines that generate the 2's complement of the input binary patterns with circuit diagram, truth table and block diagram. [4 + 6]
10. Define multiplexer. Explain 4:1 multiplexer with its block diagram, truth table and logic diagram. Implement 8:1 multiplexer with using lower order multiplexer. [2+4+4]
11. Explain how race condition in JK flipflop can be resolved? Design a 3-bit gray code synchronous counter. [4+6]

0 0 2 1